

Directed percolation and the transition to turbulence

Björn Hof  

Abstract

It may come as a surprise that a phenomenon as ubiquitous and prominent as the transition from laminar to turbulent flow has resisted combined efforts by physicists, engineers and mathematicians, and remained unresolved for almost one and a half centuries. In recent years, various studies have proposed analogies to directed percolation, a well-known universality class in statistical mechanics, which describes a non-equilibrium phase transition from a fluctuating active phase into an absorbing state. It is this unlikely relation between the multiscale, high-dimensional dynamics that signify the transition process in virtually all flows of practical relevance, and the arguably most basic non-equilibrium phase transition, that so far has mainly been the subject of model studies, which I review in this Perspective.

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Introduction

Turbulence is encountered in flows as varied as the motion of dust and gas in accretion discs, air currents past moving vehicles, water flowing from a faucet, the transport of gas and oil through pipelines or blood pulsating through the human aorta. In these examples, turbulence frequently arises, despite the fact that the laminar flow – the smooth ordered motion that persists at sufficiently low speeds – is stable to infinitesimal perturbations. Investigations of these transition types date back to the classic study of pipe flow by Reynolds¹, who famously established the sole control parameter of the problem, the Reynolds number (Re), which corresponds to the ratio of inertial to viscous forces. More precisely, Reynolds stated that turbulence can only be found if $Re = UD/\nu$ (where U is the bulk flow velocity, D is the pipe diameter and ν is the kinematic viscosity of the fluid) is large. However, Reynolds equally noted that a large Re is only a necessary but not a sufficient condition for turbulence to arise. In addition, the laminar flow must be disturbed by a perturbation of finite amplitude. Or, as Reynolds put it, “This at once suggested the idea that the condition might be one of instability for disturbance of a certain magnitude and stable for smaller disturbances”¹.

This stability to infinitesimal perturbations poses the key conceptual difficulty for this class of flows. Although linear stability theory has been successful in detecting instabilities driven by buoyancy and centrifugal forces^{2–4}, this methodology is of limited use in the case of laminar pipe and Couette flow, both of which are linearly stable up to very high (if not for all) Re ^{5,6}. In other cases, such as plane Poiseuille flow, turbulence already exists at Re far lower than that of the linear stability threshold⁵. Hence, the classic hydrodynamic stability theory can neither predict a critical point for the onset of turbulence nor provide insights into the nature of the underlying transition.

It is precisely this condition, the occurrence of transition in spite of linear stability, which has gained much attention and gave rise to an analogy to directed percolation (DP). Such a potential connection was first proposed by Pomeau⁷ for the growth of turbulent domains in a laminar flow. More specifically, Pomeau envisioned that the velocity of fronts separating turbulent from laminar flow may scale with a universal critical exponent of DP as a critical Re is surpassed.

In DP, an agent (such as water driven by gravity through a porous medium, or a virus in a population) can spread with a certain probability ($0 \leq P \leq 1$). For example, in bond percolation (Fig. 1a), the agent proliferates over time along open pores in the downward direction. One then arrives at a dynamical process in which at each discrete time step, wet sites have a probability P (probability for a pore to be open) to contaminate the neighbouring sites on the next level. For large P , that is, for high porosity, water persistently percolates (in the thermodynamical limit). However, once P falls below a critical value (in bond percolation $P_c = 0.64(2)$), the system always ends up in the dry state, that is, at some point in time, the pores connecting the remaining wet site (or sites) to the next level happen to be closed. Hence, at P_c , the system displays a phase transition to an impermeable state. The latter is also known as an absorbing state, as once entered the system remains in it for all times (wet sites cannot arise spontaneously). The phase transition is described by scaling exponents (Box 1).

In an analogous way, a turbulent flow region can spread to neighbouring laminar domains; at sufficiently low Re , it may also decay. Overall, the spatial proliferation of such turbulent domains (for example, in pipe flow)⁸ (Fig. 1b) bears similarities to the spreading of active sites in DP. To eventually arrive at a more comprehensive comparison between the two phenomena, I summarize the essential properties and mechanisms at play in transitional turbulence. This Perspective focuses on experiments, including numerical experiments, and in particular on some of the seemingly contradictory observations that have been reported in the long history of this problem.

The quest for the critical point

The finite-amplitude transition threshold for turbulence has hampered experimental investigations. The ‘natural’ transition point, that is, the value of Re at which transition first occurs in a given experiment, tends to strongly depend on experimental details such as the inflow condition; values range from $Re \approx 2,000$ – $100,000$ (ref.⁹). Even if turbulence is intentionally triggered (by adding an obstacle to the flow or via injecting a jet through a small hole in the wall, for example), depending on the type and magnitude of the disturbance, the transition point may vary considerably, with reported values of $1,750 \lesssim Re_c \lesssim 2,500$ (ref.¹⁰). Consequently, for over a century after Reynolds’ first study of

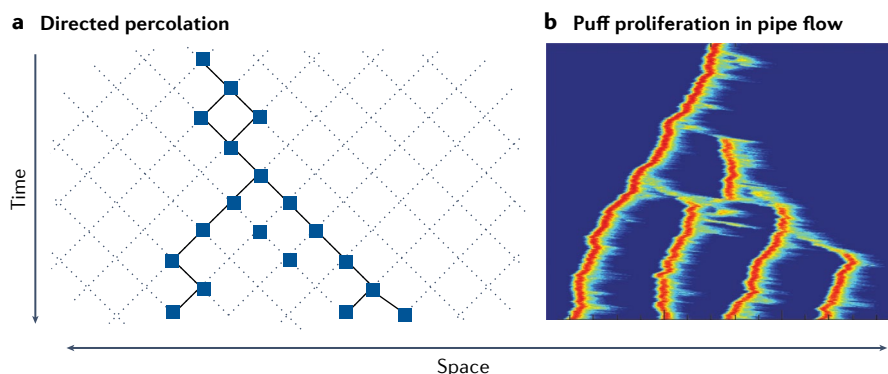


Fig. 1 | Analogy between directed percolation and proliferation of turbulent puffs. **a**, In directed percolation, an agent (such as water, a fire or an infectious disease) spreads along a preferred direction (time axis in part **a**). In this example of bond percolation, at each discrete time step, the agent spreads along open bonds (solid black lines). Active sites (blue) can hence only propagate along connected paths (originating from $t = 0$). If at any time step none of the remaining

active sites is connected to the next level, the system enters the absorbing inactive state. **b**, Space–time representation of spatiotemporal intermittency in pipe flow at Reynolds number (Re) = 2,300. Puffs can interact and decay. They can multiply by a process called puff splitting and in doing so create a second puff in the neighbouring patch of laminar fluid (blue). Part **b** adapted with permission from ref.⁸, AAAS.

the problem, experiments could not agree on a critical point for the onset of sustained turbulence.

Despite the absence of a linear instability and the practical difficulties caused by differences in experimental procedures, it is nevertheless possible to define a general critical point, that is, a point that is not specific to a given experiment or to the type of perturbations used. To do so, instead of taking the usual perspective of the laminar state, one takes the point of view of turbulence and defines the critical point as the threshold Re below which all turbulence decays. Hence, rather than looking for the destabilization of the laminar flow state, one looks for the disappearance of turbulence. Surprisingly, this definition of the critical point was already provided by Reynolds¹, who stated that “it became clear to me that if in a tube of sufficient length the water were at first admitted in a high state of disturbance, then as the water proceeded along the tube the disturbance would settle down into a steady condition, which condition would be one of eddies or steady motion, according to whether the velocity was above or below what may be called the real critical value.”

As it turns out, this brief statement precisely captures the nature of the critical point as well as the absorbing state role of the laminar flow. In many ways, this definition was far ahead of its time. The theoretical framework required to characterize absorbing-state phase transitions was unknown and theorists pursued entirely different approaches. Conversely, the simple procedure suggested by Reynolds turned out to be a virtually impossible task. The latter practical complications are essentially rooted in the spatiotemporal nature of turbulence in shear flows, which I review next, followed by a discussion of an alternative method to determine the critical point.

Localization and spatiotemporal intermittency

A property of shear flows, which was also first observed in the same paper by Reynolds for pipe flow, is that turbulence at sufficiently low Re becomes spatially localized and only appears in ‘flashes’, now more commonly referred to as puffs. In pipes and equally in square ducts or in spanwise narrow Couette setups, the resulting laminar–turbulent co-existence patterns are essentially 1D (Fig. 2a). A given puff fills the cross-section of the duct, and further growth is only possible in the streamwise direction. Conversely, in wide channels or Couette flows, at Re slightly above the relaminarization threshold, turbulence takes the form of oblique stripes (that is, tilted with respect to the main flow direction)^{11–13} and the laminar–turbulent patterns are 2D (Fig. 2b). Regardless of dimension, the flow patterns above the critical point are persistently intermittent in space and time. In other words, flow patterns continuously change as new turbulent sites arise and existing ones decay (Fig. 1b).

Similarities between spatiotemporal intermittency (STI) in transitional turbulence and STI in lower-dimensional systems such as coupled map lattices^{14–16} have frequently been pointed out. For coupled maps, it is well established that the system undergoes a phase transition to sustained STI at a specific critical point, and (for asynchronous updating) this transition has been shown to fall into the DP universality class¹⁷.

A property of transitional turbulence that is often neglected and was not included in these early model studies is the spatial interaction between neighbouring turbulent puffs^{18,19} (and similarly between stripes). As discussed subsequently, this interaction strongly affects the nature of the analogy to DP. Puffs cannot arbitrarily approach each other. Equally, two puffs or stripes cannot be combined to create a turbulent region of twice the width: turbulence automatically

Box 1

Directed percolation

The directed percolation universality class is characterized by critical exponents. Various exponents can be measured. However, many of them are related. To determine whether an absorbing state transition falls into the directed percolation universality class, it suffices to determine three independent critical exponents. The ones considered in this Perspective are: β , the scaling exponent that describes the continuous increase of the density of active (or wet) sites as P (the probability of an active site contaminating a site on the next level) is increased past the critical point; $\mu_{\perp}$, the spatial empty interval exponent; and $\mu_{\parallel}$, the temporal empty interval exponent. The latter two exponents describe the distributions of laminar intervals (in space and time, respectively) at the critical point.

shrinks back to a puff (or stripe) of the original width. For simplicity, in this Perspective, I review only the interaction and localization mechanisms for puffs in 1D flows; although it should be understood that some analogous mechanism is responsible for localization and a well-defined stripe width in 2D flows. A potential relation between the localization mechanisms responsible for puff and stripe spacing has been pointed out previously¹⁹.

Puffs require a (close to) parabolic velocity profile at their upstream interface²⁰, and at this location energy, is continuously fed into the puff and keeps it sustained. This process has been related to an inflectional instability¹⁸ (see also ref.²¹ for an alternative interpretation involving a Kelvin–Helmholtz instability). Flattening of the upstream profile has been shown to cause puffs to decay. Conversely, the profile directly downstream of a puff is plug shaped, that is, flattened, and it only gradually (with downstream distance) regains a more parabolic shape^{20,22}. This downstream development length is the cause of the minimum streamwise puff spacing, which turns out to be approximately $25D$ for pipe flow¹⁹, a value that is in agreement with stripe spacings observed in Couette flow¹⁹. Any two puffs that get closer than this distance start to interact, and the downstream puff now adjacent to a blunted rather than a parabolic profile will decay. Equally, any attempt to trigger turbulence at the downstream side of a puff is unsuccessful¹⁸.

A simple two-variable reaction–diffusion model establishes^{23,24} a remarkable analogy between puffs and excitable media. In the model, turbulence corresponds to the excited state and the parabolic flow to the excitable rest state. Incorporating the profile development reported in experiments, the downstream edge of an excitation corresponds to a refractory region across which the corresponding model variable must recover, before another excitation can be sustained. An alternative interpretation of puff localization based on an analogy to predator–prey dynamics has been presented in recent studies^{25,26}.

Note that at higher Re ($2,300 < Re < 2,800$ in pipe flow), puffs eventually begin to delocalize and turn into larger structures called slugs²². The latter transition is caused by a different physical mechanism^{23,27}, which is unrelated to DP and has been reviewed elsewhere²⁴.

Finite lifetimes and the critical point of pipe flow

In Reynolds' definition of the critical point, he cautiously stated the necessity of 'a tube of sufficient length' for the flow to settle to the respective equilibrium state. The length requirement in pipes (and channels) is caused by the advection of turbulence by the mean flow. The longer the time required by the initially perturbed flow to settle to its eventual state, the longer the tube needs to be. It is this constraint that makes a direct determination of the critical point, as suggested by Reynolds, virtually impossible. Puffs, although seemingly sustained²⁸, are transient and suddenly decay after often excessively long times. This feature applies not only to puffs in pipe flow but also equally holds for other shear flows, and the memoryless decay of turbulence has been subject to a large number of studies^{29–37}. In all of these cases, lifetimes of individual turbulent patches were found to rapidly increase with Re . Initial studies have suggested a divergence of lifetimes at a finite Re threshold^{30,31,33}, potentially defining the sought-after critical point. However, more detailed experiments^{32,35} increased the observation times by several orders of magnitude and have determined that although lifetimes increase superexponentially, they remain finite. Hence, the transient nature of puffs persists. Superexponential lifetime scalings have been confirmed in subsequent experiments^{34–36,38} for several flows as well as in direct numerical simulations^{39–42}, and theoretical arguments have been put forward, relating them to supertransients⁴³ and to rare events^{44–46} as well as to DP in finite-size cylindrical domains⁴⁷.

Going beyond individual puffs or stripes, the spatiotemporal nature of flows has been stressed and it has been argued that spatial proliferation can give rise to sustained turbulence even if individual constituents (that is, turbulent puffs or stripes) remain transient⁴⁸. Following this line of argument, it has been pointed out⁴⁹ that in the case of pipe flow, the proliferation process is the creation of new puffs

(that is, not the expansion of existing ones), a process known as puff splitting^{22,28,50}. It has been suggested that puff splitting only sets in at the critical point and gives rise to sustained turbulence (for an estimated value of $Re_c = 2,300$). Moreover, it has been noted that this transition is distinct from the onset of fully turbulent flows.

Experimentally, the puff splitting process has been quantified and its nature has been shown to be memoryless⁸, that is, the probability of a puff to seed a new one is constant in time and does not increase with the age of the puff. Puff splitting was shown to not define the critical point on its own; instead, what matters is the competition between splitting and decay. The critical point can then be approximated as the balance point at which the probability for a puff to split is identical to its probability to decay, a condition that is met at $Re \approx 2,040$ (Fig. 3). It is important to note that in general this balance point does not coincide with the critical point and the two can substantially deviate from each other (for instance, in bond percolation $P_c \neq 0.5$). In the present case, however, lifetimes and splitting times scale superexponentially with Re . Consequently, a small increase in Re beyond the balance point leads to a dominance of contamination and hence, in the thermodynamic limit, puff turbulence becomes sustained. By contrast, slightly below 2,040, the decay of puffs by far outweighs any splittings that may occur and hence turbulence overall decays. Following this line of argument, the critical point of pipe flow can be estimated to be within $\Delta Re \approx 10$ of the balance point of decay and contamination and hence $Re_c \approx 2,040 \pm 10$.

The DP analogy

Having reviewed some of the key characteristics of the onset of turbulence, we can now proceed with a more comprehensive comparison with DP. Central to this analogy is that in each of the cases considered, the laminar flow is linearly stable and acts as an absorbing state.

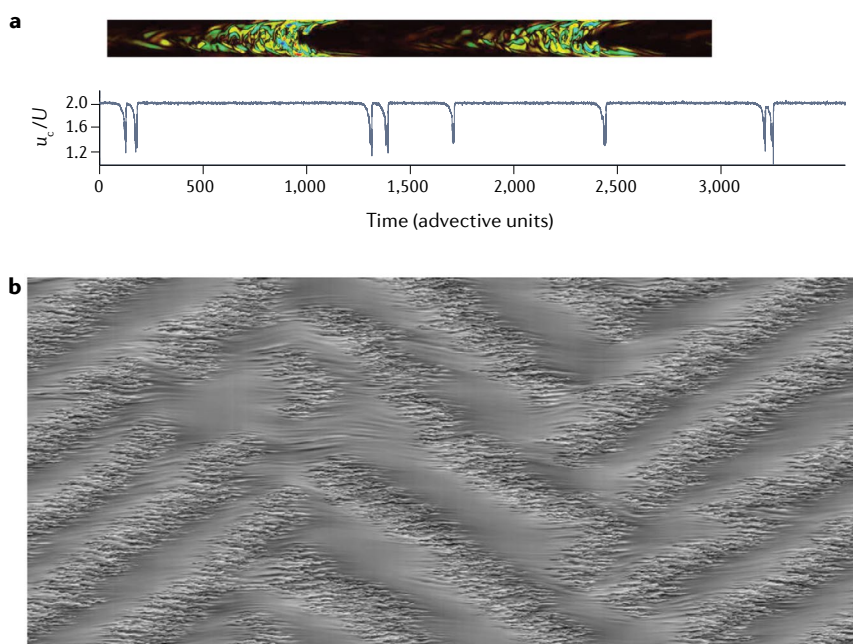


Fig. 2 | Laminar–turbulent co-existence patterns. a, Time series of the centre-line velocity (u_c) in a pipe flow experiment. In laminar (Hagen–Poiseuille) flow, u_c is twice the bulk speed (U) and a dip in the centre-line velocity corresponds to the location of a turbulent puff. Two representative puffs are shown from direct numerical simulations (colours correspond to different streamwise vorticity

levels, flow from right to left). **b**, Spatiotemporal intermittency in Couette flow. Criss-crossing turbulent stripes are interspersed by more quiescent, quasi-laminar regions. Part **a** adapted with permission from ref.¹⁰, Cambridge Univ. Press. Part **b** adapted with permission from ref.⁶¹, Cambridge Univ. Press.

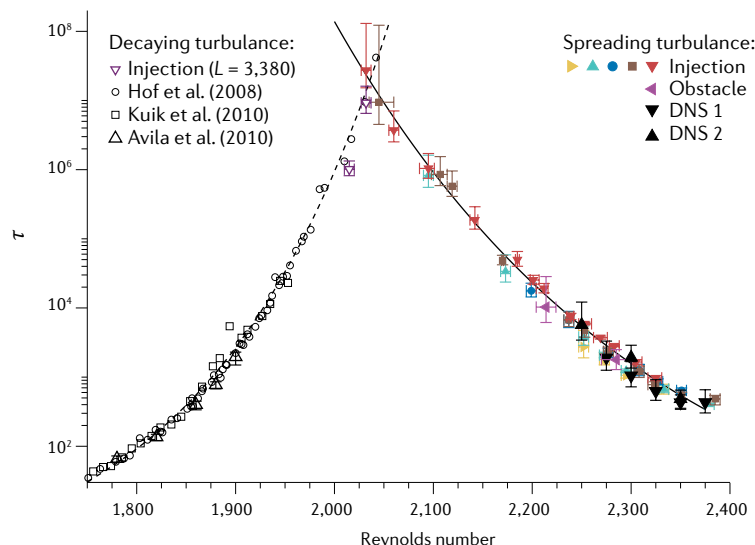


Fig. 3 | Median lifetimes (left branch) and splitting times (right branch) of isolated turbulent puffs in pipe flow. Lifetimes and splitting times scale superexponentially with Reynolds number (Re) and intersect at $Re = 2,040$. Hence, for $Re > 2,040$, puffs proliferate faster than they decay and, owing to the rapid change of the curve with Re , the intersection point provides a close approximation of the critical onset of sustained turbulence. Whereas most data were measured experimentally, the lifetime data in Ref.³⁷ were obtained in direct numerical simulations (DNS), so were the two data sets for puff splitting denoted by DNS1 and DNS2. The error bars denote 95% confidence intervals. Figure adapted with permission from ref.⁸, AAAS.

The existence of a unique absorbing state is one condition of what is known as the Janssen–Grassberger DP conjecture^{51,52}, a list of four requirements for a model to fall into this universality class. Although the absorbing-state property was also the basis of the proposition by Pomeau⁷, he had a laminar–turbulent interface in mind, whereas the analogy I describe here takes macroscopic puffs and stripes as individual active sites. A crucial requirement for this refined analogy is the fluctuating (in the sense of non-persistent) nature of the active phase, which in the present case is connected to the transient memory-less nature of puffs^{32,35}. If puffs were to become sustained at a critical point as earlier studies had suggested^{30,31,33}, the transition would not belong to the DP universality class.

In addition to the continuous transition from a fluctuating into a unique absorbing state, the DP conjecture states that the transition must be characterized by a positive one-component order parameter. In this case, the order parameter is proportional to the turbulent fraction, more precisely the fraction of the flow covered by puffs (or stripes).

The third condition states that interactions must be short range (so, for example, puffs should only interact across finite distances). Whether this condition is met in shear flows is not clear a priori. In principle, pressure could give rise to long-range interactions. However, interactions for shear flows that have been reported to date are short range, even in some cases they may go beyond the next nearest neighbour¹⁰.

Last, the system must not have any special attributes such as additional symmetries or quenched disorder. The latter occurs in glasses and polymer materials, for instance. Such attributes are at least not known for transitional flows, and so this condition is probably met. For further discussions of the DP conjectures, see ref.⁵³.

A more subtle and peculiar aspect of this analogy arises from the contamination process via puff (or stripe) splitting and from the localized nature of puffs^{18,19}. Although this aspect as such is a microscopic detail and does not affect the compliance with the DP conjecture, it impacts the nature of the active phase. A puff (or stripe) in many ways should be considered as a laminar–turbulent interface or, more correctly, a pair of interfaces (upstream and downstream) that travel at the same speed^{23,27}. The contamination via puff splitting does not extend the

existing laminar–turbulent interface but it entails the creation of a new pair of laminar–turbulent interfaces at a separate downstream location.

This difference to the original DP conjecture first of all affects the temporal and spatial scales that need to be considered. In itself this is only a quantitative change, but it has dire consequences for experimentalists: taking the example of pipe flow, at $Re = 2,040$, the median time for a splitting to occur exceeds 10^7 advective time units (Fig. 3), a time interval in which puffs travel by 10^7 pipe diameters downstream. In analogy to DP, this splitting time then corresponds to a single time step in the discretized model sense. The equivalent of a moderate DP simulation with 10^3 time steps correspondingly requires a tube of length equal to 10^{10} times its diameter. It is unlikely that Reynolds had such scales in mind when he referred to ‘a tube of sufficient length’.

The second aspect of the peculiar nature of puff splitting is that the seemingly obvious identification of an active site as turbulence and of laminar flow as inactive, as has been proposed originally⁷ and as has equally been stated recently⁵⁴, is incorrect or at least incomplete. A segment of turbulence cannot survive at these Re without energy input from the upstream parabolic flow²⁰. Hence, an active site is composed of laminar as well as turbulent flow and includes the upstream parabolic segment (Fig. 4a). This equally applies to flows in two spatial dimensions, in which instead of puffs, turbulence is composed of stripes that remain localized and are surrounded by (quasi)laminar regions.

Although this difference may appear subtle, a direct consequence is that the DP phase transition cannot explain the eventual onset of space-filling turbulence. Instead, the two phases of the flow associated with this transition are the active phase composed of strictly localized puffs (or stripes with a minimum spacing) and the absorbing fully laminar flow. In a study of stripes in a (spatially 1D) Couette flow³⁹ (Fig. 4b), the flow was initiated with a single seed significantly above critical ($\varepsilon = (Re - Re_c)/Re_c = 0.1$). At this value, the splitting (that is, contamination) rate outweighs the decay probability by more than two orders of magnitude. It is therefore safe to assume that this flow is entirely dominated by splittings and hence equivalent to a contamination probability close to unity in DP. The flow shows the conical spread of the active states as expected for a single seed experiment in this limit; however, the resulting flow is not fully turbulent but consists

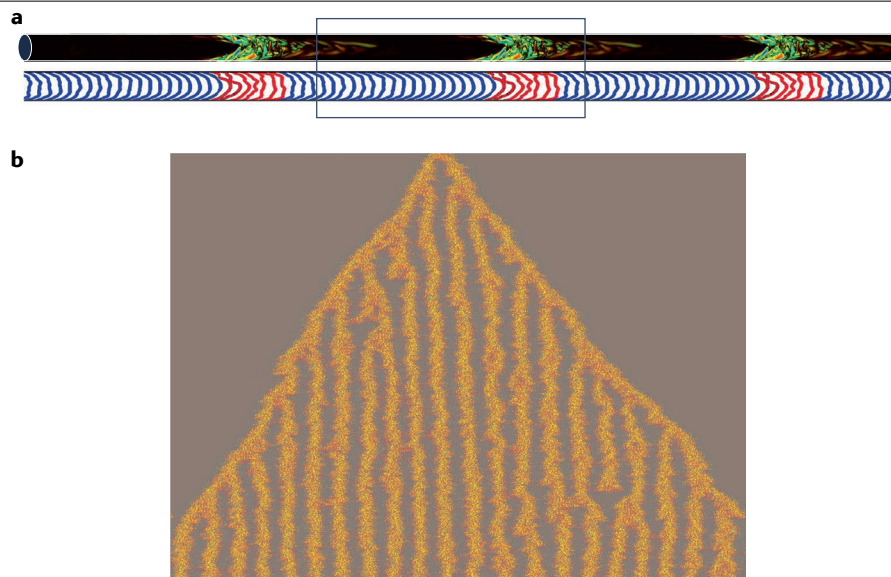


Fig. 4 | Definitions of active sites and the order parameter. **a**, Densely spaced active sites in the pipe flow. The black box indicates one active site. Regions of high vorticity (top row) and the corresponding velocity profiles (bottom row), with high turbulent kinetic energy (red) and relaminarizing regions (blue), across which the velocity profile recovers towards a more parabolic shape, are shown. **b**, Space–time plot of a ‘single seed’ experiment in a direct numerical simulation

of a quasi-1D Couette flow. At the final time, puffs are densely seeded, corresponding to a fully occupied lattice and an order parameter of 1, although the turbulent fraction is <1 . The order parameter is hence proportional to the turbulent fraction but not equal to it. Part **a** courtesy of Baofang Song. Part **b** adapted with permission from ref.³⁹, APS.

of precisely spaced stripes that (except for short transient intervals) adhere to the minimum spacing requirement outlined earlier. Likewise, in 2D extended flows, a fully occupied lattice does not correspond to fully turbulent flows but to densely packed stripes that adhere to some minimum spacing⁵⁵ (Fig. 2b). Such dense stripe patterns should hence be considered the equivalent of the $P=1$ limit in DP. In this interpretation, dense patterns arise from local interactions and are hence self-organized. This is in contrast to earlier interpretations⁵⁶ of stripe patterns, which have suggested a global instability of turbulence to be responsible for the alternating laminar turbulent regions.

At this point, a remark on terminology is in order, specifically regarding the frequent referral to this transition as ‘subcritical’. ‘Critical’ in this case implies a linear instability of the laminar state and it is hence inferred that the origin of turbulence is connected to such an instability, which may or may not exist at higher Re . By contrast, it is now well established⁵⁷ that turbulence arises from exact coherent structures that are entirely unrelated to the stability of the laminar flow (exact coherent structures originate in saddle-node bifurcations). What causes further confusion is that the term ‘subcritical’ implies a discontinuous transition, whereas the DP phase transition is in actual fact continuous. The important distinction here is between the continuity (or discontinuity) of a (disturbance) amplitude in the bifurcation context and that of the order parameter of a phase transition (that is, the fraction of flow covered by puffs or stripes). The amplitude of an individual turbulent structure (such as a puff before decay) remains finite and does not approach the laminar level at the critical point. The phase transition can nevertheless be continuous.

To illustrate this distinction with an example, consider an epidemic outbreak. As the effective reproduction number approaches zero at the end of the epidemic, any remaining infected individual will still have

the typical (finite) viral load, just like the turbulent kinetic energy of an intact puff remains non-zero. The quantity that approaches zero continuously is rather the fraction of the population that is infected.

On a separate note, even if the phase transition is continuous, this concept only holds in the thermodynamic limit. For a finite systems size, such as a pipe of finite length, an apparent discontinuity of the order parameter is observed. In pipes, once the typical spacing between puffs approaches the pipe length, the flow will eventually relaminarize, even though Re is above critical.

Quantitative evidence for DP

Pomeau’s idea was initially tested in several experimental studies^{29,58–60}. However, these studies remained mostly inconclusive. In particular, for the case of planar Couette flow, experiments have reported a discontinuous (with respect to the order parameter) transition^{29,58}, a result that was later also obtained in direct numerical simulations⁶¹. These studies appeared to end the speculation regarding a possible connection to DP. In retrospect, and in the light of the more recent developments⁸ outlined earlier, the domain sizes in those earlier studies were at least an order of magnitude too small in each direction, circumstances under which aforementioned finite-size effects lead to an apparent discontinuity.

Following the introduction of the adjusted DP analogy, numerous recent experiments and direct numerical simulations have proposed continuous transitions and DP exponents for various shear flows, including (circular) Couette^{55,62}, channel⁶³, boundary layer⁶⁴, Kolmogorov⁶⁵, Waleffe⁶⁶ and annular Couette flow⁶⁷. Generally, such measurements must resolve all relevant temporal and spatial scales in the vicinity of the critical point. Flow patterns become scale invariant in this limit, and to determine the respective scaling exponent, domain sizes are required that exceed the minimum puff (or stripe)

spacing – that is, the size of an active site (Fig. 4a) – by significantly more than an order of magnitude (note that this requirement is not met by all aforementioned studies). Measurements can only commence after all initial transients have died out, so that flow patterns reach a statistical steady state, a timescale typically much longer than the turbulence proliferation time (for instance, in pipes the timescale is equal to multiple puff-splitting times). Finally, to obtain sufficient scaling ranges for the temporal exponents, these already very large timescales must be exceeded by yet another order of magnitude or more.

Compared with these challenges, the actual measurement techniques required to experimentally obtain quantities of interest, such as the turbulent fraction or the size distributions of laminar gaps, are straightforward. Typically, the fluid is seeded with tracer particles, and recordings are carried out with high-speed cameras. Figure 5 summarizes the scaling of the order parameter and the distributions of the laminar gap sizes in space and time, determined in aforementioned flows. For comparison, the expected DP exponents are plotted alongside. Although other critical exponents have also been tested in these studies, the three shown are the most frequently measured. Moreover, these three exponents are independent of each other and hence sufficient to test whether a given flow falls into the DP universality class.

Figure 5a–c displays the results of those studies proposing agreement with DP in 1 + 1D. A reasonable scaling range and close agreement with 1 + 1D DP are obtained for the experiments in circular Couette flow. Note that those data were accompanied by direct numerical simulations, which are also in good qualitative agreement, although owing to the smaller domain sizes used, the scaling ranges are smaller. Compared with pipes and channels, circular Couette flow has the big advantage of periodic boundary conditions, and hence observation times are not limited by the domain size. The problem was further simplified by choosing a small axial (spanwise) aspect ratio and a very large azimuthal (streamwise) aspect ratio. In such a domain, flow patterns are essentially 1D and consist of short stripe segments analogous to puffs in pipe flow. Therefore, equally, the contamination only occurs in the streamwise direction. Also in the computational study of Kolmogorov⁶⁵ flow, a 1D extended domain was chosen. The data are consistent with DP, although the scaling ranges (except for $\mu_{||}$, the exponent of distribution of the laminar gap sizes in time) are typically less than an order of magnitude. The annular Couette simulations⁶⁷ focused on a novel regime that occurs for azimuthal sizes $\gg 2\pi$, a regime that hence cannot be realized in experiments. In this case, flow patterns are clearly 2D, yet it appears that the contamination predominantly occurred via splitting in the azimuthal direction. In addition, owing to the large computational costs, scaling ranges are moderate; nevertheless, the results are consistent with the DP analogy proposed earlier.

Finally, 1 + 1D DP exponents have also been suggested for a boundary layer flow⁶⁴. However, in boundary layers, turbulent structures proliferate in two spatial dimensions and it is not a priori clear why 1D exponents should be expected. Moreover, the control parameter (that is, Re) is not constant in space but increases in the streamwise direction, which is also the direction in which turbulent structures travel. In doing so, they contaminate the downstream flow. Nevertheless, data were taken focusing only on the spanwise contamination and the flow was visualized at fixed streamwise locations corresponding to the fixed values of Re. The scaling ranges are typically smaller than a decade (Fig. 5a–c) and hence the data remain inconclusive. Overall, the transition in this flow follows different rules and is difficult to compare with the scenario described earlier.

I next look at the transition in cases in which STI occurs in two spatial dimensions, that is, flows that at least in the thermodynamic

limit are infinitely extended not only in the streamwise but also in the spanwise direction. An experimental study of channel flow⁶³ first proposed 2 + 1D DP. The scaling ranges obtained are, however, only moderate (Fig. 5d–f). A key problem is that, owing to the mean advection of turbulence and the short streamwise domain length, observation times are limited to approximately 2,000 advective time units. This timescale is many orders of magnitude shorter than those of stripe decay and stripe splitting in channel flow⁴¹. The authors of the experimental study suggested to avoid the extremely long timescales in the following way: “To overcome this difficulty, we chose a quasi-two dimensional channel flow and forced the inlet boundary condition to be an active (turbulent) state”⁶³. Specifically, the inlet flow was set to be fully turbulent, which, as discussed earlier, does not correspond to the active state in the spatiotemporally intermittent regime. The first non-trivial state that becomes sustained in channel flow consists of strictly localized stripes^{13,68–71}. Conversely, fully turbulent initial conditions correspond to a state that is not stable at these Re. In pipes, attempts to enforce a fully turbulent flow in the intermittent regime counterintuitively cause premature relaminarization^{18,27,72} and consequently an overestimate of the critical point. Indeed, the transition point of $Re = 830$ suggested in ref.⁶³ for channel flow disagrees with various studies, which reported that turbulent stripes are sustained to $Re < 700$ (refs.^{68,73,74}). The most recent estimate for the critical point of the absorbing-state phase transition in channel flow⁷¹ is $Re_c \approx 650$.

It should also be noted that in the vicinity of the critical point, fully localized turbulent stripes appear to have some unexpected properties. In computational studies⁷³ in which patterns were influenced by periodic boundary conditions, channel flow stripes were reported to all have the same orientation. Conversely, in experiments of larger domain size, both orientations can be found. However, stripes of opposing orientation have opposite lateral drift speeds and therefore separate. At the same time, new stripes created by stripe splitting keep the orientation of their parent stripe and hence overall patterns of opposing orientation segregate (assuming an infinite domain). Another marked difference to other shear flows is that channel stripes are not memoryless⁷¹, a property probably connected to a deterministic growth process that results from an instability at the leading tip of stripes⁷⁵. Owing to these unusual properties, it is unclear whether channel flow belongs to the DP universality class. Either way, further studies of this case are needed, but at the same time, measurements in the vicinity of the critical point require aspect ratios many orders of magnitude larger than the largest experiments performed to date.

More conclusive than channel flow has been a numerical study conducted for the Waleffe model⁶⁶ (a simplification of Couette flow with stress-free boundary conditions). Heavily under-resolving the simulations made it possible to realize very large aspect ratios. Overall, the transition was found to be in excellent agreement with DP in 2 + 1D, as testified by the critical exponents (Fig. 5d–f).

The critical exponents for 2 + 1D DP were finally also confirmed in an experimental investigation⁵⁵ of 2D stripe patterns in a circular Couette flow, with a far larger axial aspect ratio (1,000 gap width) when compared with that of ref.⁶². In addition, these two Couette flow studies have demonstrated that the DP transition is thus robust for this flow and applies to 1D and 2D flow patterns.

Discussion

Although many of the studies mentioned earlier found exponents that are consistent with DP, often the scaling ranges are only moderate. In this context, it is important to recall that to safely deduce a power law

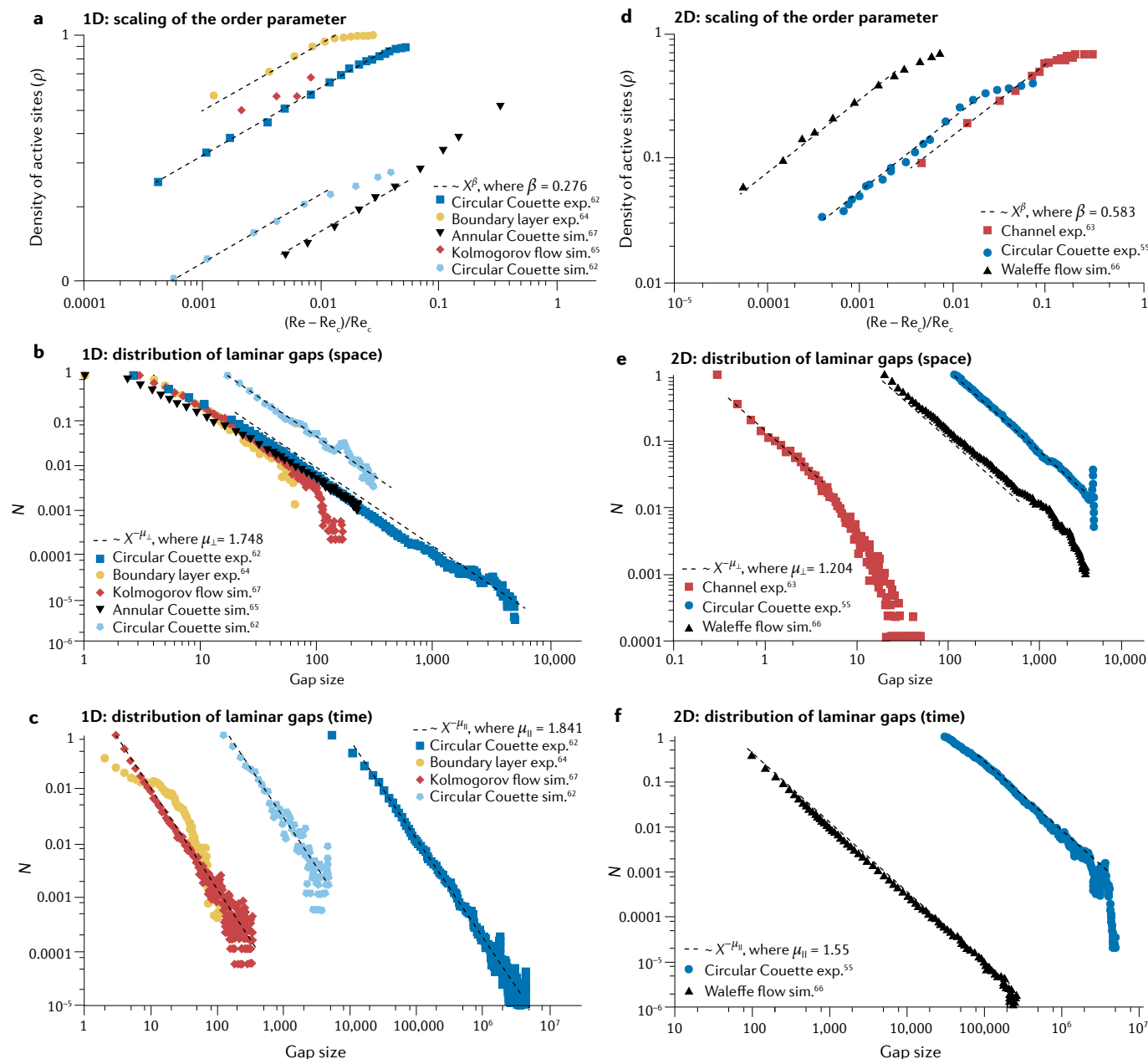


Fig. 5 | Comparison of measurements of critical exponents from recent years. **a–c**, Critical exponents (exp.) for 1D flows, that is, flows in which turbulent sites (puffs) can only proliferate in one spatial dimension. Data taken from refs.^{62,64,65,67}. Circular Couette simulations (sim.) are direct numerical simulations. **d–f**, 2D flows in which proliferation takes place in two spatial dimensions. Data taken from refs.^{55,63,66}. Parts **a** and **d** show data for the scaling of the order parameter, that is, the density of active sites (ρ). The corresponding exponent in 1D directed percolation is 0.276 and that in 2D is 0.583 (lines in parts **a** and **d**, respectively).

Parts **b** and **e** show the distribution of the laminar gaps, which follows a power law sufficiently close to the critical point (that is, the critical Reynolds number (Re_c)). The corresponding critical exponent, the so-called spatial empty gap exponent $\mu_\perp$ (lines), is plotted alongside the data sets from the different experiments. The numerical value of the exponent is $\mu_\perp = 1.748$ for 1D and $\mu_\perp = 1.204$ for 2D. Parts **c** and **f** show the distribution of the laminar gaps in time. The universal scaling with exponent $\mu_\parallel$ is indicated by the lines; the numerical value for 1D is $\mu_\parallel = 1.841$ and for 2D is $\mu_\parallel = 1.55$. β , scaling exponent.

from data, one should aim for two decades⁷⁶ in scaling range. Many of the aforementioned cases only show a decade or less and thus cannot provide a conclusive answer. Probably most delicate are the fits for the β exponents because to fit for β requires also to fit the critical point. Even a small

shift in Re_c will strongly affect β . In particular, there are several cases (the study of channel flow⁶³, the study of a boundary layer separation bubble⁶⁴ and, to a lesser extent, also the study of Kolmogorov flow⁶⁵) in which the order parameter does not go to zero below the suggested critical point.

Such a gradual, rounded onset cannot be accounted for by finite-size effects. As discussed earlier, finite-size effects lead to a premature relaminarization into the absorbing state and hence to a discontinuity and an overestimate of the critical point. A rounding below the critical point occurs if the measurement time is insufficient and the flow has not yet reached its statistical steady state. There is a simple remedy for this problem: the observation time needs to be extended (however, in a frame of reference that moves with the turbulent structures) to determine whether the points in question eventually reach the absorbing state. However, doing so is not always possible in experiments.

The danger in disregarding non-zero data points is easily demonstrated by considering a classic measurement by Rotta⁷⁷ of the turbulent fraction for pipe flow (Fig. 6a). I selected the left-most curve (the one recorded furthest from the inlet) and truncated the data in one case at $Re_{tr} = 2,375$ and in the other at $Re_{tr} = 2,419$. This simple exercise results in two entirely different apparent power laws (Fig. 6b): one with an exponent close to 0.5 and the other close to 0.276. In both cases, the data follow the supposed scaling for close to a decade. Of course, in reality, the data do not follow a power law at all and neither of the selected truncation values corresponds to the critical point for pipe flow. In this case, a longer observation time would have shown that the truncated data points in actual fact remain non-zero and that overall the data have not reached the statistical steady state, a point that Rotta was fully aware of. A rounding below an actual critical point would shrink as the observation time is increased; for the example in Fig. 6, instead the curves keep shifting to the left with time (Fig. 6a) (time here is proportional to the downstream position x/d).

It is exactly this property of shear flows, the extremely slow approach to the statistical steady state, that poses a considerable challenge for future studies. Given the non-equilibrium nature of the phenomenon and given that the contamination and other microscopic processes vary between different flows, there is no guarantee that the critical behaviour found in one geometry can be transferred to another. For pipe flow, despite its historical importance, the direct experimental

classification of the transition will probably remain out of reach. Even though in principle the memoryless nature of puffs can be exploited to effectively create periodic boundary conditions in experiments¹⁰ and timescales of 10^8 advective time units have been reached, there were no signs of scale invariance and it is probable that a much closer approach to the critical point is needed and probably timescales in excess of 10^{10} advective time units would have to be resolved. Such an undertaking would take decades of physical measurement time. Likewise, in direct numerical simulations, computational costs would be excessive.

However, there are various model results that I did not review here, such as the briefly mentioned excitable media^{23,24} and predator–prey^{25,26} models, which do suggest that pipe flow should fall into the DP universality class. Because direct evidence appears unobtainable, experimentally inspired models can serve as an intermediate step. Such models would not only incorporate qualitative features but also the quantitative microscopic interactions measured in the actual flow problem. Yet, another possibility is to explore closely related geometries such as curved pipes or pipes of non-circular cross-section, to see whether in these geometries the contamination timescales shift to lower values.

Despite these formidable problems, the transition to turbulence – at least in geometries with periodic boundary conditions such as circular Couette flow – provides one of the few experimental realizations of DP. Including the numerically approachable cases such as Waleffe, Kolmogorov and annular Couette flow, various systems are available already (and hopefully more will follow) to test the DP conjecture and the robustness to changes in features such as boundary conditions or fluid properties. It will be particularly interesting to see whether other DP analogies, for example, one closer to the original proposition by Pomeau of the small-scale dynamics at a laminar–turbulent interface, may also be found in other situations. In particular, a formidable challenge in this respect is set by jet flows and boundary layers in which parameters vary spatially.

So, does DP finally explain the nature of the transition to turbulence? This question is to some extent ill-posed, because depending on the context, the term transition can relate to different aspects of the

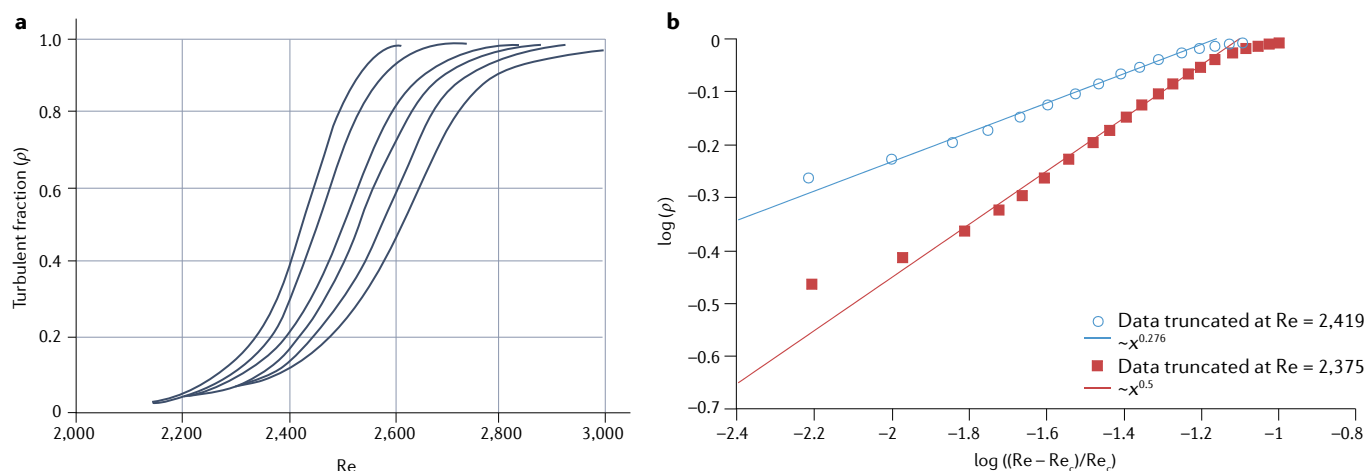


Fig. 6 | Dangers in disregarding non-zero data points. **a**, Turbulent fraction (ρ ; also known as intermittency factor) as a function of Reynolds number (Re) measured at six different distances from the pipe inlet. The curves shift to the left with downstream distance, indicating that ρ has not reached its statistical steady state. All curves have a characteristic S-shape and do not show any indication of criticality, such as a discontinuity in the derivative that would be expected for the order parameter at a second-order phase transition. **b**, If one assumes that

the upper part of the curve measured furthest from the inlet has reached the statistical steady state and sets a critical point, ignoring the non-zero data below, a seeming scaling range of close to a decade with an exponent of 0.276 can be observed for the order parameter (blue data). However, for a different choice of the critical point, a comparable ‘scaling range’ with an entirely different exponent is obtained. Part **a** adapted with permission from ref. ⁷⁷, Springer.

overall scenario. The classic ‘laminar perspective’ and the associated question of whether a given perturbation causes a flow to become turbulent are entirely outside the scope of the DP framework, as are questions relating to the origin of chaos, or how turbulent transients arise from invariant solutions, and so on.

However, it is in many ways the virtue of the statistical mechanics framework that it can identify universal properties irrespective of the overall complexity of the problem and its microscopic details. It is noteworthy that the critical point associated with this non-equilibrium phase transition is the only one that can be reasonably and precisely defined for transition, although it corresponds to the reverse transition to the absorbing laminar flow. One of the most remarkable aspects of the long history of transition studies and the quest for critical points, involving luminaries such as Kelvin, Rayleigh, Sommerfeld and Heisenberg⁷⁸, is that the absorbing-state nature of the transition had already been clearly pointed out by Reynolds in the very first study of the problem, yet it took over a century to fully appreciate its implications.

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Correspondence should be addressed to Björn Hof.

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